



Effect of the Fear Factor and Prey Refuge in an Asymmetric Predator–Prey Model

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Abstract

This study investigates the influence of fear, refuge, and migration in a predator–prey model, where the interactions between the species follow an asymmetric function response. In contrast to some other findings, we propose that prey develop an anti-predator response in response to a concentration of predators, which in turn increases the fear factor of the predators. The conditions under which all ecologically meaningful equilibrium points exist are discussed in detail. The local and global dynamics of the model are determined at all equilibrium points. The model admits several interesting results by changing the rate of fear of predators and predator aggregate sensitivity. Numerical simulations have been performed to verify our theoretical findings. We found that under certain conditions, the system appears to be losing the stability to acquire the periodic attractor when the trait-mediated direct effect of one of each (prey growth, competition, fear, and prey death).

Keywords Predator–prey model · Fear · Refuge · Additional food · Migration · Bifurcation

1 Introduction

Mathematical modeling helps with the analysis of natural phenomena in a concise and better way. In the literature, differential equations and dynamical systems have been used

for modeling many physical phenomena that occur in physics [1], ecology [2–4], biology [5–7], and other applied sciences [8]. One of these phenomena is the relation between prey and predator, the foundation of which was put by Lotka and Volterra, and the basic result they concluded is the periodic behavior of prey and predator populations [9–12, 33]. Many research works have already been done on various mathematical models that describe the relation between prey and predator [13–17]. Naik et al. [18] examined the effect of the strong allee effect on the predator–prey system and observed the complex dynamics, including different kinds of bifurcations. In [19], the effect of the fractional-order derivative and predator harvesting rate on prey–predator dynamics has been discussed.

The common belief in predator–prey relation is that predators only impact prey numbers through killing. But there is another opinion that has emerged recently, which is that apart from killing, the presence of the predator alone (i.e., fear of predation) leads to a great deal of behavioral and physiological changes among the prey group, which reduces the fertility and reproduction of the prey and its survival. On the other hand, some authors also studied the effect of prey refuge on the dynamics of the predator–prey system. The authors in [20] studied the effect of refuge in a prey–predator model. The refuge impact on prey with a Holling type II is suggested in [21]. Wang et al. [22] discussed the effect of refuge on a

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prey–predator system. On the other hand, some authors of [23–28] present a mathematical model for refuge impact. In [29], the authors investigated the fear and refuge effects of prey on predator–prey dynamics, suggesting that both factors play a vital role in stabilizing the system. Pal et al. [30] discussed how the hunting cooperation of predators and the anti-predator behavior of prey influence the dynamics of a predator–prey Leslie-Gower model.

The influence of the fear effect and prey refuge on prey–predator dynamics was studied in [31], which suggested that increasing fear levels can improve the system's stability. Chen et al. by [32] investigated the impact of predator fear on prey species. Zhu et al. [33] studied the influence of fear on the Lotka-Volterra predator–prey system. In [34], it was reported that the number of predators can be reduced by increasing fear and the Allee factor. Wen et al. [35] performed the bifurcation analysis of a predator–prey system and explored the role of the Allee factor and anti-predator behavior. The fear effect induced by immature predators and prey refuge on prey–predator dynamics via a fractional-order delay stage structure model has been discussed in [36]. Das and Samanta [37] studied the impact of additional food and predator fear on prey species. In [38], the authors explored a square root predator–prey model with the fear of prey. Thus, the literature proves that the fear of prey and prey refuge plays a vital role in stabilizing prey–predator dynamics. On the other hand, the meaning of migration is the relatively long-distance movement of individual animals. It is found in all major animal groups, including birds, mammals, fish, reptiles, amphibians, insects, and crustaceans. The cause of migration may be local climate, local availability of food, the season of the year, or mating [39]. There are many researchers who have studied the migrant effect on the dynamics of ecological models such as [40] which proposed a mathematical model of the prey–predator system with the effect of migrating and infectious disease. In [41], the authors explored a predator–prey model under the immigration effect. Also, Bergam et al. [42] studied the influence of density-dependent migrations on the dynamics of a predator–prey system. In [43], the author discussed the effect of immigration and refuge on the predator–prey model. In [44, 45], the authors formulated a mathematical model of a prey–predator with migration. Therefore, in this study, we develop a predator–prey model to examine the influence of fear, refuge, and migration in a prey–predator system where the interactions between the species follow an asymmetric function response.

The order of this paper is as follows: Sect. 2 is devoted to the formulation of the model. The existence of equilibria is in Sect. 3. Section 4 discusses the mathematical analysis to establish local and global stability results for the proposed model. Some applications of our main results are presented

through numerical simulation in Sect. 5. Eventually, the discussion of the finding results and concluding remarks are given in Sect. 6.

2 Model Formulation

In this part, the prey–predator model is formulated mathematically using an asymmetric function response to describe the model. The model has two non-linear ordinary differential equations describing how the population densities of the two species would vary with time. We formulate our model based on the following assumptions:

- H1 : The prey population grows by the logistic function in the absence of the predator.
- H2 : The presence of predators induces fear in the prey, leading to a decrease in the prey's birth rate due to cognitive stress, while it can increase the death rate of prey.
- H3 : The functional response of the predator to the prey is assumed to be of asymmetric type.
- H4 : One of the natural factors is the migration of a predator when there is a shortage of food (prey or an alternative food source). Thus, the model equations and the flow chart of the model are given below:

$$\begin{aligned}\frac{dx}{dt} &= x \left[\frac{r}{1+fy} - \frac{\beta(1-m)y}{\psi y + \alpha\theta + (1-m)x} - a(1+fy) - \kappa x \right], \quad x(0) \geq 0 \\ \frac{dy}{dt} &= y \left[\frac{e\beta(1-m)x}{\psi y + \alpha\theta + (1-m)x} - d - \frac{\sigma}{(1-m)x + \alpha\theta} - \epsilon y \right], \quad y(0) \geq 0.\end{aligned}\quad (2.1)$$

Here, $x(t)$ and $y(t)$ represent the densities at time t for the prey and predator, respectively (Fig. 1). Then, based on the above assumptions, the parameters of our model are written in Table 1.

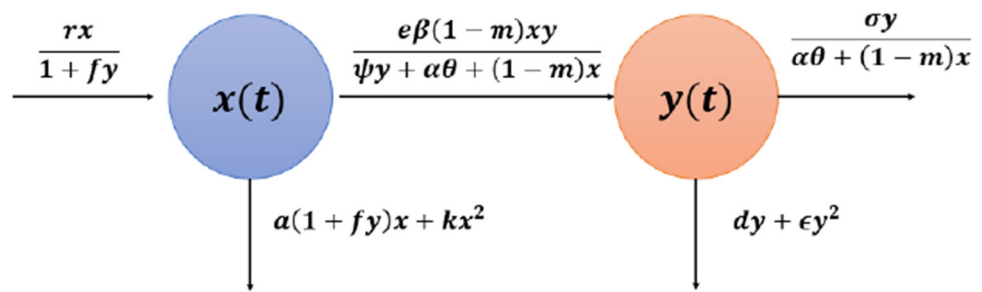
3 Equilibria

In this part, the presence of all the equilibrium points (EPs) is calculated as follows:

- The model (2.1) has an extinction equilibrium point (EEP) $W_0 = (0, 0)$.
- The predator-free equilibrium point (PFEP), which is denoted by $W_1 = (x_1, 0)$, exists uniquely under $a < r$ which is the biological condition where

$$x_1 = \frac{r-a}{k}. \quad (3.1)$$

Fig. 1 Diagram of prey–predator model



- There exists a positive interior equilibrium point (PIEP) $W_2 = (x_2, y_2)$, where x_2 and y_2 are the positive root intersection points of the two isoclines:

$$\begin{aligned} g_1(x, y) = & r(\psi y + \alpha\theta) + r(1-m)x \\ & - \beta(1-m)(1+fy)y - a(\psi y + \alpha\theta) \\ & - a(1-m)x - k(1+fy)(\psi y + \alpha\theta)x \\ & - k(1-m)(1+fy)x^2 = 0. \end{aligned} \quad (3.2)$$

$$\begin{aligned} g_2(x, y) = & e\beta\alpha\theta(1-m)x + e\beta(1-m)^2x^2 \\ & - \sigma[\alpha\theta + (1-m)x] - \sigma\psi y \\ & - d[\alpha\theta + (1-m)x]^2 - d\psi[\alpha\theta + (1-m)x]y \\ & - \epsilon[\alpha\theta + (1-m)x]^2y - \epsilon\psi[\alpha\theta + (1-m)x]y^2 = 0. \end{aligned} \quad (3.3)$$

Next, when $y \rightarrow 0$, we have the following results:

$$g_1(x) = A_1x^2 + A_2x + A_3, \quad (3.4)$$

where

$$\begin{aligned} A_1 = & -k(1-m) < 0, \\ A_2 = & [(r-a)(1-m) - k\alpha\theta], \\ A_3 = & \alpha\theta(r-a). \end{aligned} \quad (3.5)$$

Table 1 Definitions of model parameters

Parameter	Biological meaning
r	The growth rate of prey
β	The attack rate
e	The conversion rate from prey to predator,
κ and ϵ	The competition rates
f	The fear rate
m	The refuge rate of the prey
$\alpha\theta$	The quality and quantity of additional food
a and d	The death rates of population
σ	The migration rate
ψ	The half-saturation constant
$x(0)$ and $y(0)$	The initial point

$$g_2(x) = B_1x^2 + B_2x + B_3, \quad (3.6)$$

where

$$\begin{aligned} B_1 = & (e\beta - d)(1-m)^2, \\ B_2 = & e\beta\alpha\theta(1-m) - \sigma(\alpha\theta + (1-m)) - 2d\alpha\theta(1-m), \\ B_3 = & -d\alpha^2\theta^2 < 0. \end{aligned} \quad (3.7)$$

So, (3.4) has a unique positive root under the biological condition, and the value of $\tilde{x}_2$ is as follows:

$$\tilde{x}_2 = \frac{-A_2 - \sqrt{A_2^2 - 4A_1A_3}}{2A_1}.$$

Again, (3.6) has a unique positive root when $d < e\beta$ and the value of $\bar{x}_2$ are as follows:

$$\bar{x}_2 = \frac{-B_2 + \sqrt{B_2^2 - 4B_1B_3}}{2B_1}.$$

Clearly, the two isoclines $g_1(x, y) = 0$ and $g_2(x, y) = 0$ have a unique intersection point (x_2, y_2) , provided that under the following conditions (see Fig. 2.)

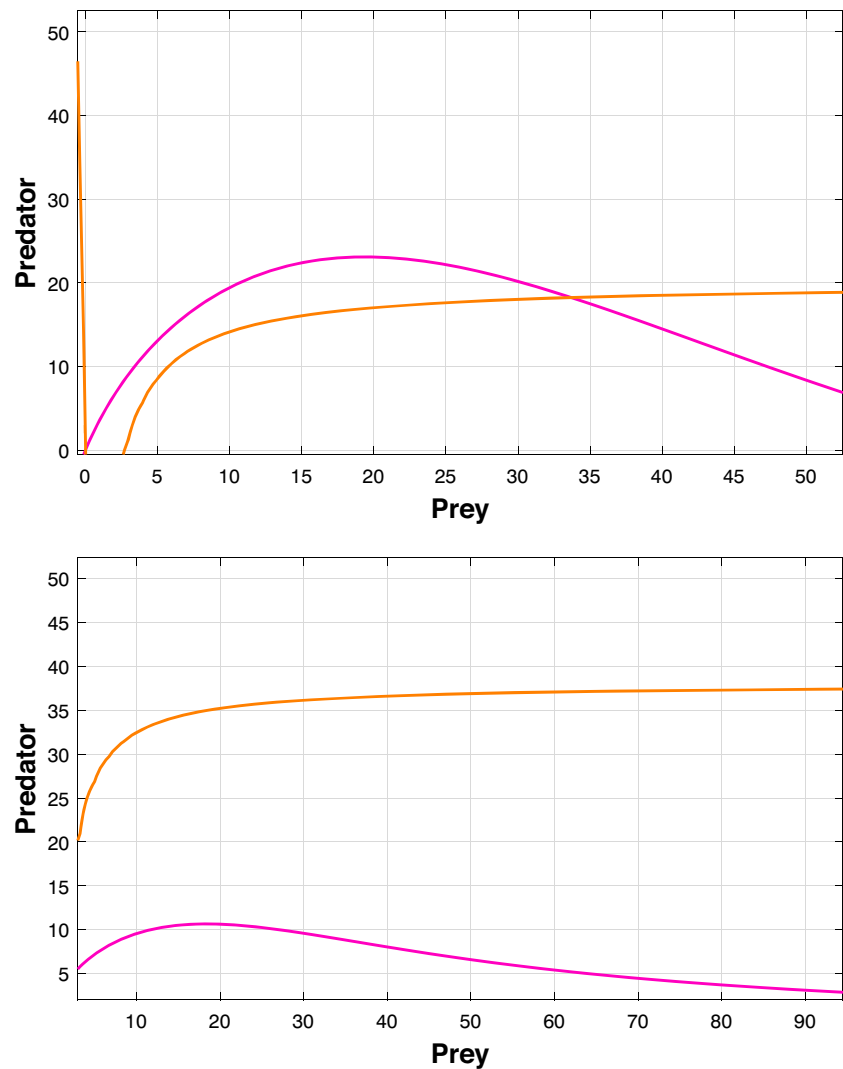
$$-\frac{\partial g_1(x, y)/\partial x}{\partial g_1(x, y)/\partial y} < 0,$$

$$-\frac{\partial g_2(x, y)/\partial x}{\partial g_2(x, y)/\partial y} > 0.$$

4 Stability

In this part, we will analyze the equilibria in terms of their local stability using eigenvalues to identify conditions that can help understand the dynamical behavior of model (2.1). We apply the trace-determinant theorem. Let $W = (x, y)$ be an arbitrary equilibrium point of the model (2.1). Hence, the

Fig. 2 Isocline diagram of the model (2.1) around positive interior equilibrium point W_2 (magenta color curve for the prey isocline and orange color curve for the predator isocline)



associated characteristic equation of the Jacobian matrix of the proposed model at W is given by the following:

$$J_W = \begin{pmatrix} \frac{r}{1+fy} - \frac{T_1}{T^*} - a(1+fy) - 2\kappa x & \frac{-rfx}{(1+fy)^2} - \frac{xT_2}{T^*} - afx \\ \frac{eT_1}{T^*} - \frac{\sigma(1-m)y}{[(1-m)x+\alpha\theta]^2} & \frac{exT_2}{T^*} - d - \epsilon - \frac{\sigma}{[(1-m)x+\alpha\theta]^2} \end{pmatrix}. \quad (4.1)$$

Here,

$$T_1 = \beta(1-m)(\psi y + \alpha\theta)y,$$

$$T_2 = \beta(1-m)(\alpha\theta + (1-m)x),$$

$$T^* = [\psi y + \alpha\theta + (1-m)x]^2.$$

Theorem 4.1 The extinction equilibrium point (EEP) is a locally asymptotically point if the biological condition is not satisfied (i.e., $r < a$); otherwise, it is unstable.

Proof Rewriting the (4.1) about W_0 , we get the following results:

$$J_{W_0} = \begin{pmatrix} r-a & 0 \\ 0 & -\left(d + \epsilon + \frac{\sigma}{(\alpha\theta)^2}\right) \end{pmatrix}. \quad (4.2)$$

Clearly, the diagonal Jacobian matrix of model (2.1) at W_0 has two eigenvalues: $\lambda_x = r - a$ and $\lambda_y = -\left(d + \epsilon + \frac{\sigma}{(\alpha\theta)^2}\right) < 0$. Then, we have the first eigenvalue negative under the condition $r < a$. Hence, the (EEP) point is a locally asymptotically equilibrium point. $\square$

Theorem 4.2 The predator-free equilibrium point (PFEP) is locally asymptotically stable if all the biological conditions $a < r$ hold, and the following conditions are met:

$$ex_1\beta(1-m)(\alpha\theta + (1-m)x_1) < \sigma. \quad (4.3)$$

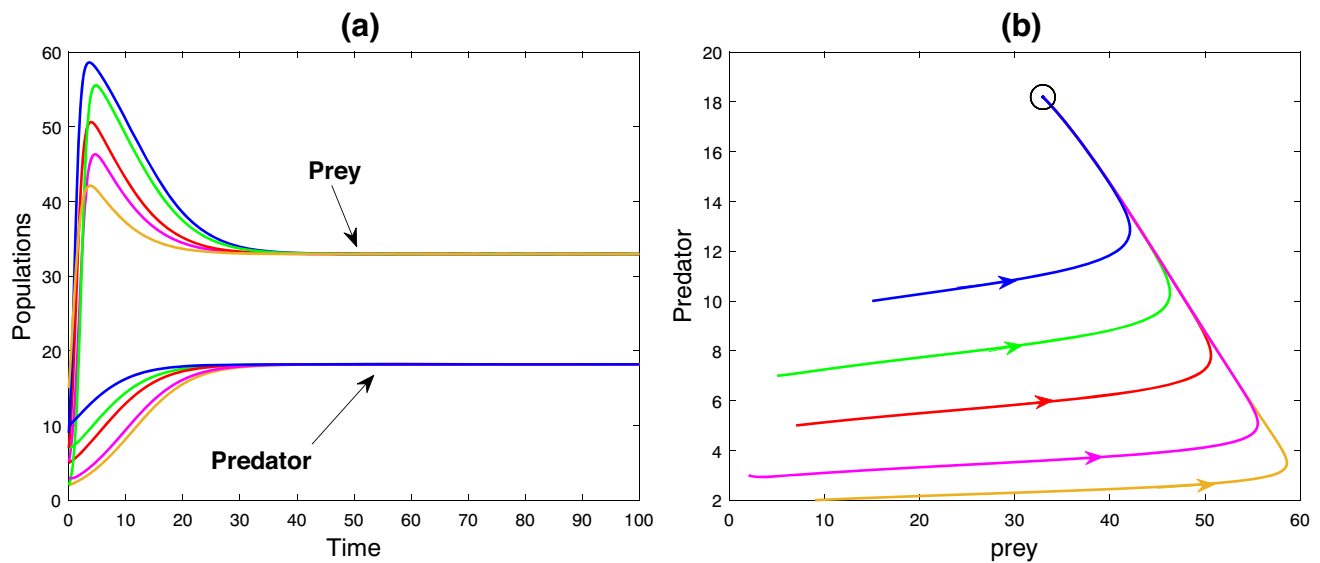


Fig. 3 The graphical representation of model (2.1) according to (6.1) by different initial points converge to $W_2 = (33, 18)$: **a** time series of trajectories and **b** 2D phase plot

Proof Evaluating the (4.1) at W_1 and get a new formulation

$$J_{W_1} = \begin{pmatrix} -(r-a) & -\left(rf + af + \frac{\beta(1-m)}{\alpha\theta + (1-m)x_1}\right)x_1 \\ 0 & -d - \epsilon - \frac{\sigma}{[\alpha\theta + (1-m)x_1]^2} + \frac{ex_1\beta(1-m)}{\alpha\theta + (1-m)x_1} \end{pmatrix}. \quad (4.4)$$

Thus, the Jacobian matrix around W_1 has two eigenvalues, namely λ_x and λ_y . These eigenvalues have to be negative

when the conditions (4.3) and the biological state ($a < r$) are met. Then, the (PFE P) point is a locally asymptotically stable point and a saddle point otherwise.

Theorem 4.3 The positive equilibrium point (PIEP) is locally asymptotically point stable if and only if the following conditions are met:

$$\frac{r}{1+fy_2} < \alpha(1+fy_2) + 2\kappa x_2 + \frac{\beta(1-m)(\psi y_2 + \alpha\theta)y_2}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2}, \quad (4.5)$$

$$\frac{\sigma(1-m)}{(\psi y_2 + \alpha\theta)[(1-m)x_2 + \alpha\theta]^2} < \frac{e\beta(1-m)}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} < \frac{(d+\epsilon)[(1-m)x_2 + \alpha\theta]^2 + \sigma}{x_2[(1-m)x_2 + \alpha\theta]^3}.$$

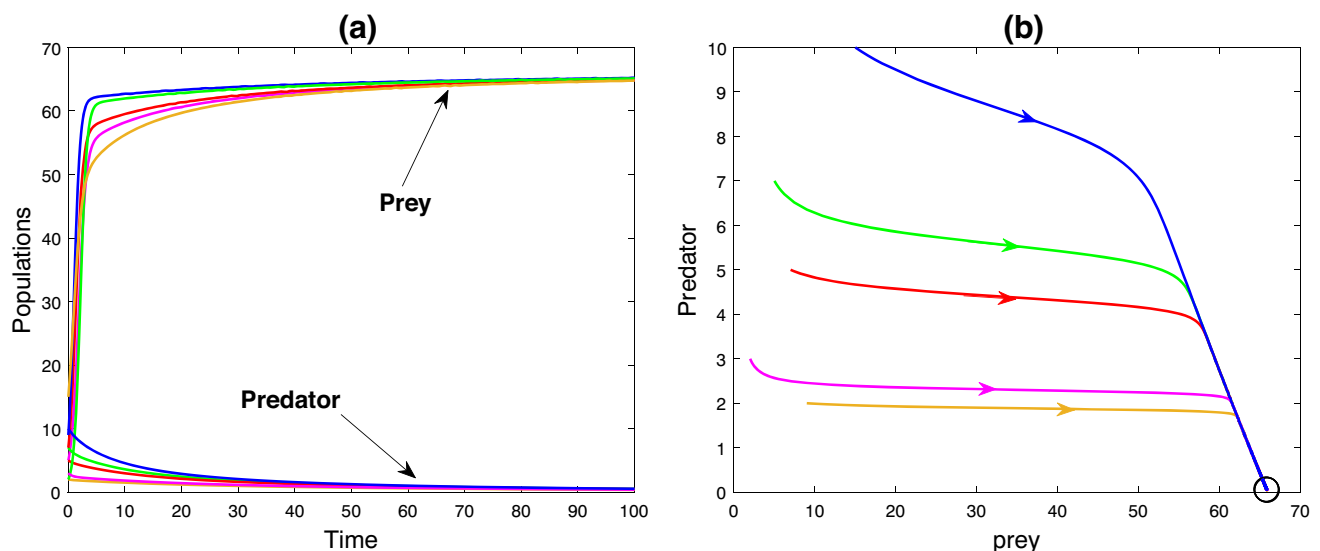


Fig. 4 The graphical representation of model (2.1) according to (6.1) with $d = 0.2$ and different initial points converge to $W_1 = (66, 0)$: **a** time series of trajectories and **b** 2D phase plot

Proof Evaluating the (4.1) at W_2 , we get a new formulation

$$J_{W_2} = \begin{pmatrix} \xi_{11} & \xi_{12} \\ \xi_{21} & \xi_{22} \end{pmatrix}. \quad (4.6)$$

Here,

$$\begin{aligned} \xi_{11} &= \frac{r}{1+fy_2} - \frac{\beta(1-m)(\psi y_2 + \alpha\theta)y_2}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} - \alpha(1+fy_2) - 2\kappa x_2, \\ \xi_{12} &= \frac{-rfx_2}{(1+fy_2)^2} - \frac{\beta x_2(1-m)(\alpha\theta + (1-m)x_2)}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} - \alpha f x_2, \\ \xi_{21} &= \frac{e\beta(1-m)(\psi y_2 + \alpha\theta)y_2}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} - \frac{\sigma(1-m)y_2}{[(1-m)x_2 + \alpha\theta]^2}, \\ \xi_{22} &= \frac{e\beta x_2(1-m)(\alpha\theta + (1-m)x_2)}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} - d - \kappa - \frac{\sigma}{[(1-m)x_2 + \alpha\theta]^2}. \end{aligned}$$

Clearly, from the above formula, we get the characteristic equation

$$\lambda^2 + \eta_1 \lambda + \eta_2 = 0, \quad (4.7)$$

where

$$\begin{aligned} \eta_1 &= -(\xi_{11} + \xi_{22}), \\ \eta_2 &= \xi_{11}\xi_{22} - \xi_{12}\xi_{21}. \end{aligned}$$

Thus, the (PIEP) equilibrium point (W_2) is locally asymptotically stable under the conditions (4.5) and an unstable point otherwise. $\square$

Next, we will discuss the global stability and determine the basin of attraction for all equilibrium points by finding the favorable conditions to determine the behavior of our model.

Theorem 4.4 *The extinction equilibrium point (EEP) is globally asymptotically stable if the condition below is satisfied:*

$$\sigma < \alpha^2 \theta f ((1-m) + \alpha\theta). \quad (4.8)$$

Proof We define the Lyapunov function as follows:

$$\mathcal{F}_1 = x(t) + y(t). \quad (4.9)$$

Calculating the time derivative of $\mathcal{F}_1$ along the positive solution of model (2.1), we have the following summarize:

$$\begin{aligned} \frac{d\mathcal{F}_1}{dt} &= \frac{\sigma}{\alpha\theta(1-m) + \alpha\theta} - \frac{m\sigma}{\alpha\theta(1-m) + \alpha\theta} \\ &\quad - \frac{\beta(1-m)(1-e)}{\psi + \alpha\theta + (1-m)} - \frac{rf}{1+fy} - \alpha f - \kappa x^2 - \epsilon y^2. \end{aligned}$$

According to (4.8), we get $\frac{d\mathcal{F}_1}{dt} < 0$ is negative. Thus, the (EEP) is globally asymptotically stable, and the proof is complete. $\square$

Theorem 4.5 *The predator-free equilibrium point (PFEP) is globally asymptotically stable if the condition below is satisfied:*

$$\begin{aligned} 1 &< x_1, \\ (1-m)(1-x_1) &< \psi. \end{aligned} \quad (4.10)$$

Proof We define the Lyapunov function as follows:

$$\mathcal{F}_2 = y(t) = y \left[\frac{e\beta x(1-m)}{\psi y + \alpha\theta + (1-m)x} - \frac{\sigma}{(1-m)x + \alpha\theta} - d - \epsilon y \right]. \quad (4.11)$$

Calculating the time derivative of $\mathcal{F}_2$ along the positive solution of model (2.1), we have

$$\begin{aligned} \frac{d\mathcal{F}_2}{dt} &= \frac{\sigma xy(1-m)(1-x_1)}{[(1-m)x + \alpha\theta][(1-m)x_1 + \alpha\theta]} \\ &\quad - e\beta xy(1-m) \left(\frac{x(1-m)(1-x_1) + \psi y}{[\psi y + \alpha\theta + (1-m)x][\alpha\theta + (1-m)x_1]} \right) - \epsilon y^2. \end{aligned}$$

According to (4.10), we get $\frac{d\mathcal{F}_2}{dt} < 0$ is negative. Thus, the (PFEP) is globally asymptotically stable, and the proof is complete. $\square$

Theorem 4.6 *The positive equilibrium point (PIEP) is globally asymptotically stable if the condition below is satisfied:*

$$\begin{aligned} \beta y_2(1-m)^2 &< \kappa Q_1, \\ \mathcal{P}_{12}^2 &< 4\mathcal{P}_{11}\mathcal{P}_{22}. \end{aligned} \quad (4.12)$$

Proof We define the Lyapunov function as follows:

$$\mathcal{F}_3 = \left(x - x_2 - x_2 \ln \frac{x}{x_2} \right) + \left(y - y_2 - y_2 \ln \frac{y}{y_2} \right). \quad (4.13)$$

Calculating the time derivative of $\mathcal{F}_3$ along the positive solution of model (2.1), we have the following summarize:

$$\mathcal{F}_3 = \left(\frac{x - x_2}{x} \right) \frac{dx}{dt} + \left(\frac{y - y_2}{y} \right) \frac{dy}{dt}.$$

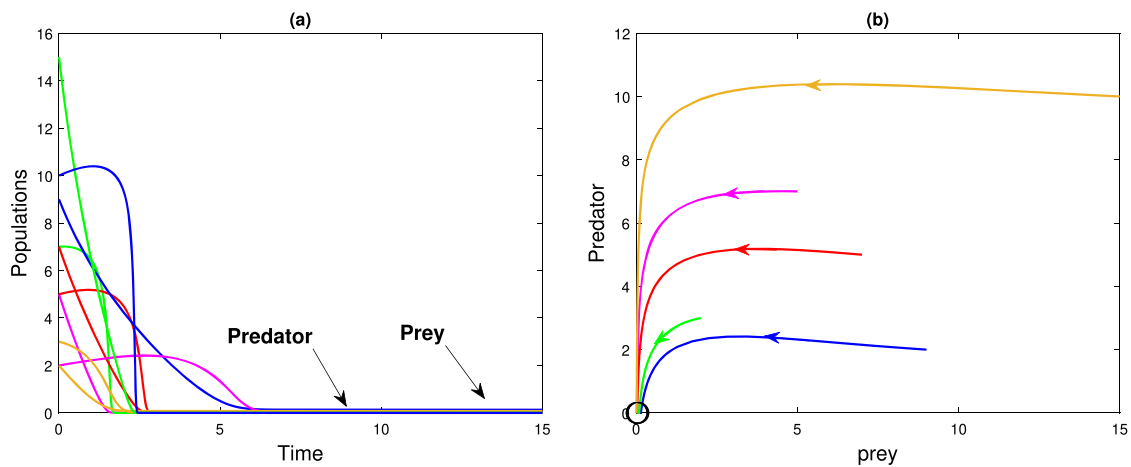


Fig. 5 The graphical representation of model (2.1) according to (6.1) with $r = 0.002$ and different initial points converge to $W_0 = (0, 0)$: **a** time series of trajectories and **b** 2D phase plot

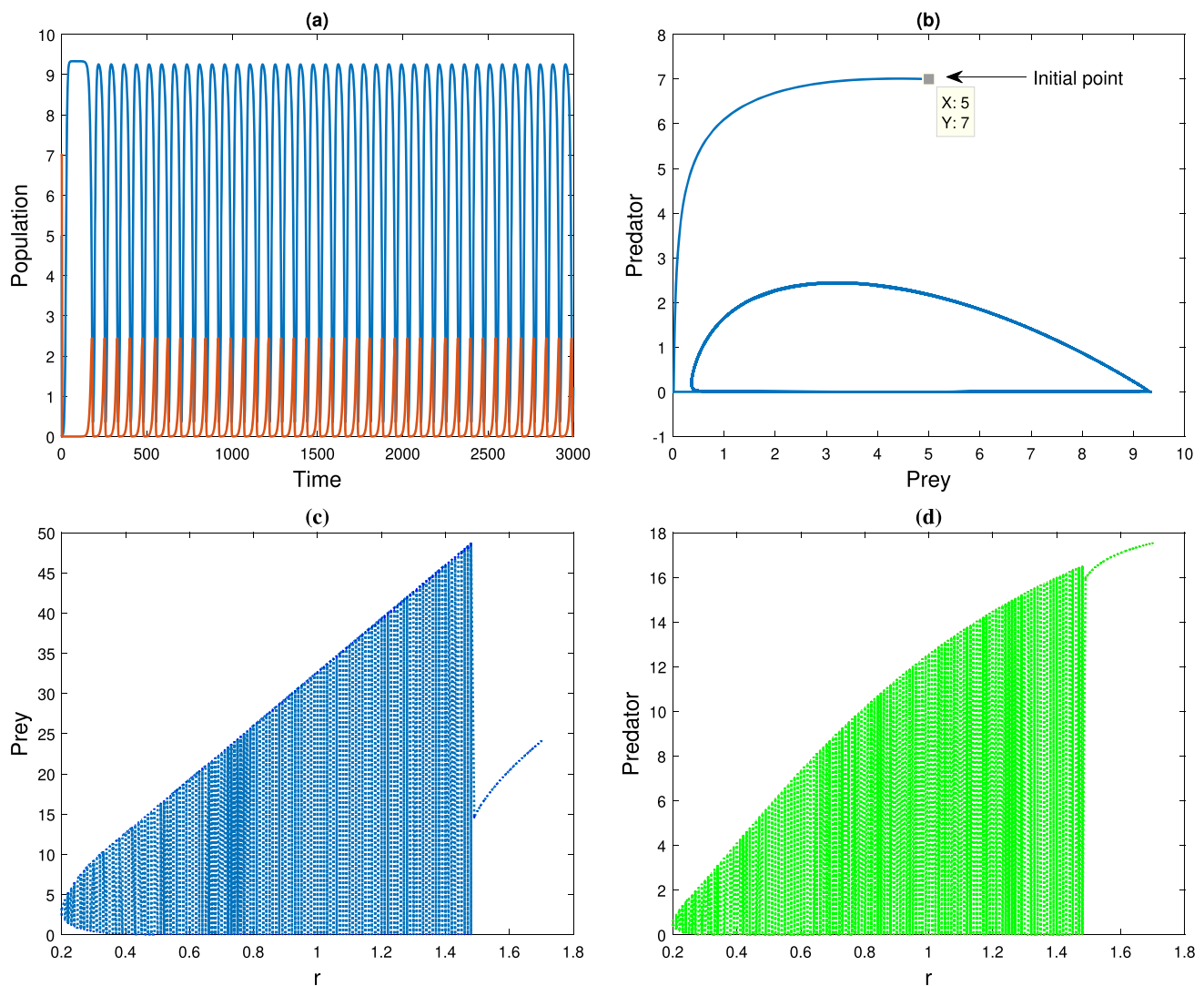


Fig. 6 The graphical representation of model (2.1) according to (6.1) with $r = 1.2$: **a** Periodic attractor. **b** 2D periodic attractor. **c** The bifurcation diagram of prey as a function of r . **d** The bifurcation diagram of the predator as a function of r

By simplifying this equation, we get the following:

$$\frac{d\mathcal{F}_3}{dt} = -[\mathcal{P}_{11}(x - x_2)^2 - \mathcal{P}_{12}(x - x_2)(y - y_2) + \mathcal{P}_{22}(y - y_2)^2],$$

Here,

$$\begin{aligned}\mathcal{P}_{11} &= \kappa - \frac{\beta y_2(1-m)^2}{Q_1}, \\ \mathcal{P}_{22} &= \epsilon + \frac{e\beta\psi x_2(1-m)}{Q_1}, \\ \mathcal{P}_{12} &= \alpha f + \frac{e\beta(1-m)(\psi y_2 + \alpha\theta)}{Q_1} + \frac{\sigma(1-m)}{Q_2} \\ &\quad - \frac{f}{Q_3} - \frac{\beta(1-m)}{Q_1}(\alpha\theta + x_2(1-m)), \\ Q_1 &= [\psi + \alpha\theta + (1-m)][\psi y_2 + \alpha\theta + (1-m)x_2], \\ Q_2 &= [(1-m) + \alpha\theta][(1-m)x_2 + \alpha\theta], \\ Q_3 &= 1 + f(1 + y_2 + fy_2).\end{aligned}$$

Clearly, according to (4.10), we have

$$\frac{d\mathcal{F}_3}{dt} \leq -\left[\sqrt{\mathcal{P}_{11}}(x - x_2) - \sqrt{\mathcal{P}_{22}}(y - y_2)\right]^2.$$

Thus, $\frac{d\mathcal{F}_3}{dt} < 0$ is negative; the (PIEP) is globally asymptotically stable, and hence, the proof is complete. $\square$

5 Hopf Bifurcation Analysis

In this part, one may discuss the occurrence of hopf bifurcation in the model (2.1) about the coexistence equilibrium point. A Hopf bifurcation arises in the model (2.1) when a stable equilibrium undergoes a transition to sustained oscillations as a parameter passes through a critical value. This bifurcation is characterized by the appearance of complex conjugate eigenvalues with zero real parts in the linearized system.

Theorem 5.1 *If the bifurcation parameter r exceeds a critical point, the model (2.1) at the positive equilibrium point (PIEP) approaches Hopf bifurcation. At $r = r^*$, the Hopf bifurcation conditions arise:*

$$\begin{aligned}\eta_1 &= 0, \\ r^* &= (1 - fy_2) \left[d + k(1 + 2x_2) + \alpha(1 + fy_2) + \frac{\sigma}{[(1-m)x_2 + \alpha\theta]^2} \right. \\ &\quad \left. + \frac{(1-m)\beta[\psi y_2^2 + \alpha\theta y_2 - ex_2(\alpha\theta + (1-m)x_2)]}{[\psi y_2 + \alpha\theta + (1-m)x_2]^2} \right], \\ 1 &> fy_2,\end{aligned}$$

$$\begin{aligned}(\psi y_2 + \alpha\theta)y_2 &> ex_2(\alpha\theta + (1-m)x_2), \\ \frac{d}{dr}(\eta_1)_{r=r^*} &\neq 0.\end{aligned}\tag{5.1}$$

Proof At the threshold system parameter value $r = r^*$, the characteristic (4.7) may be represented as

$$\lambda^2 + \eta_2 = 0.\tag{5.2}$$

It is clear that for the threshold system parameter value $r = r^*$, the characteristic equation has a pair of purely imaginary eigenvalues as $\lambda_{1,2} = \pm i\sqrt{\eta_2(r)}$. This is a characteristic feature of Hopf bifurcation.

Additionally, we need to verify the transversality condition, which states that $\frac{d}{dr}(\eta_1)$ should be nonzero. In this case, $\frac{d}{dr}(\eta_1) = \frac{-1}{1+fy_2} \neq 0$, which confirms the satisfaction of the transversality condition. $\square$

6 Numerical Simulation

We perform numerical simulations with the help of PPLANE8 and MATLAB software. The numerical simulations show various dynamics of the model (2.1) and also confirm our theoretical findings. The following set of biologically feasible hypothetical parameter values is used:

$$\begin{aligned}r &= 2, \beta = 0.5, \kappa = 0.03, \epsilon = 0.01, f = 0.03, m = 0.1, \alpha = 0.00003, \\ \theta &= 0.00002, a = 0.02, d = 0.00002, \sigma = 0.5, \psi = 0.01, e = 0.4.\end{aligned}\tag{6.1}$$

This data set shows the global stability of the positive point and the results of Theorem 4.6 (see Fig. 3).

Clearly, Fig. 3 shows that the system (2.1) is globally asymptotically stable at (PIEP).

Now, to confirm the result of Theorem 4.5, we used the same data set in (6.1) with $d = 0.2$. Thus, the solution of model (2.1) approaches the predator-free equilibrium point (PFEP) (see Fig. 4).

Obviously, Fig. 4 shows that the system (2.1) is globally asymptotically stable at (PFEP).

Also, keeping all parameter values of (6.1) but changing $r = 0.002$, we get the solutions of model (2.1) converge to the globally asymptotically stable extinction equilibrium point (EEP) (see Fig. 5).

According to other Figs. 6, 7, 8, 9, and 10, the impact of the values of the parameters such as r, f, a, κ, e , and ϵ controls the stability of the model (2.1). It is observed that the system converges in 2D periodically when $r = 1.2$ (see Fig. 6).

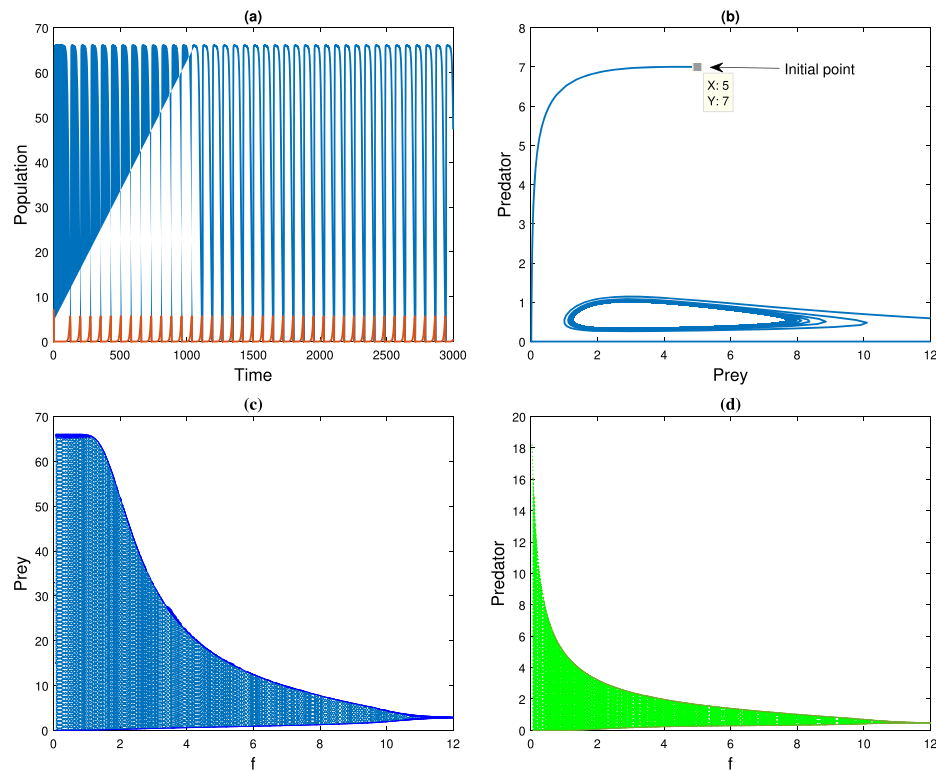


Fig. 7 The graphical representation of the model (2.1) according to (6.1) with $f = 0.5$: **a** Periodic attractor. **b** 2D periodic attractor. **c** The bifurcation diagram of prey as a function of f . **d** The bifurcation diagram of the predator as a function of f

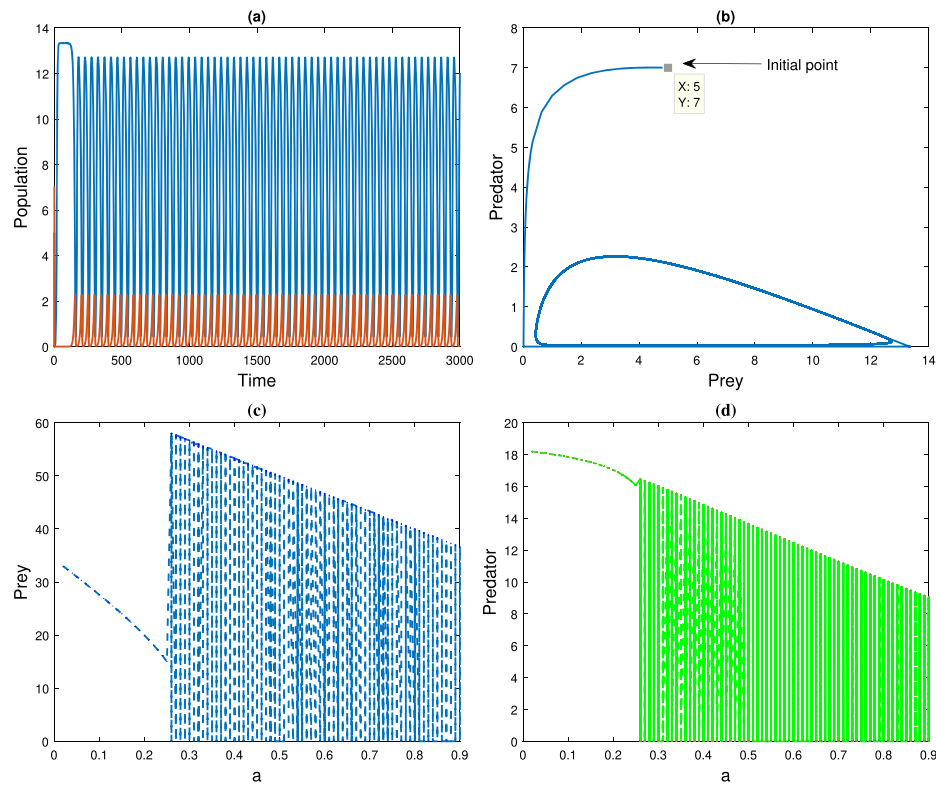


Fig. 8 The graphical representation of the model (2.1) according to (6.1) with $a = 0.4$: **a** Periodic attractor. **b** 2D periodic attractor. **c** The bifurcation diagram of prey as a function of a . **d** The bifurcation diagram of the predator as a function of a

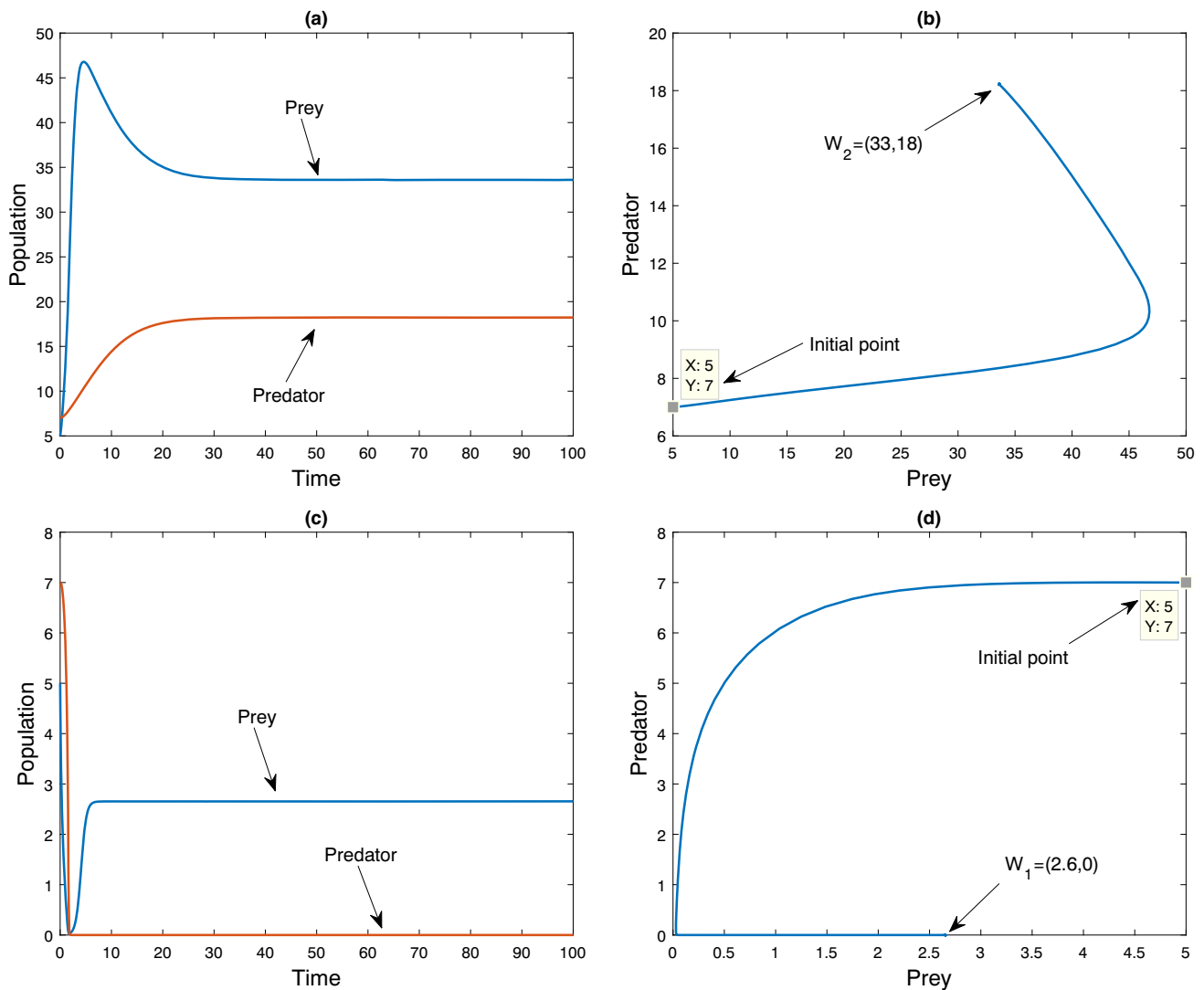


Fig. 9 The graphical representation of model (2.1) according to (6.1) with different values of κ : **a** Time series of trajectories of W_2 with $\kappa = 0.03$. **b** 2D phase plot of W_2 with $\kappa = 0.03$. **c** Time series of trajectories of W_1 with $\kappa = 0.0003$. **d** 2D phase plot of W_1 with $\kappa = 0.0003$

Figure 7 shows the impact of the increasing of the fear factor f in prey and losing the stability of W_2 .

If the prey death rate a is increasing due to fear from predators then the dynamical behavior of the model (2.1) converges to the periodic attractor of (PIEP), (see Fig. 8).

Now, it is found that the model (2.1) approaches to W_1 when the decreasing of the competition rate for prey κ , while converges to 2D periodic attractor to W_2 point when the decreasing of the competition rate for predator ϵ . See Figs. 9 and 10, respectively.

It has been found that the model (2.1) approaches W_1 when the conversion rate $e \leq 0.01$. Conversely, the model converges to W_2 when the conversion rate is $0.01 < e \leq 0.5$. This means that the dynamics of model (2.1) lose stability at W_1 and shift to stability at W_2 . However, if the conversion

rate $e \geq 0.6$, the dynamics of model (2.1) lose stability at W_2 and approach a 2D periodic attractor near the W_2 point (see Fig. 11).

Now, we demonstrate the impact of the refuge for the prey on the model dynamics based on the parameter values in (6.1) with different values for the refuge parameter $m = 0.1, 0.5, 0.9$. Figure 12 shows the results we obtained, indicating that the model's behavior remains stable towards the positive interior equilibrium point (PIEP) W_2

7 Discussion and Conclusion

We proposed a mathematical model to forecast the prey–predator dynamic with an asymmetric function response.

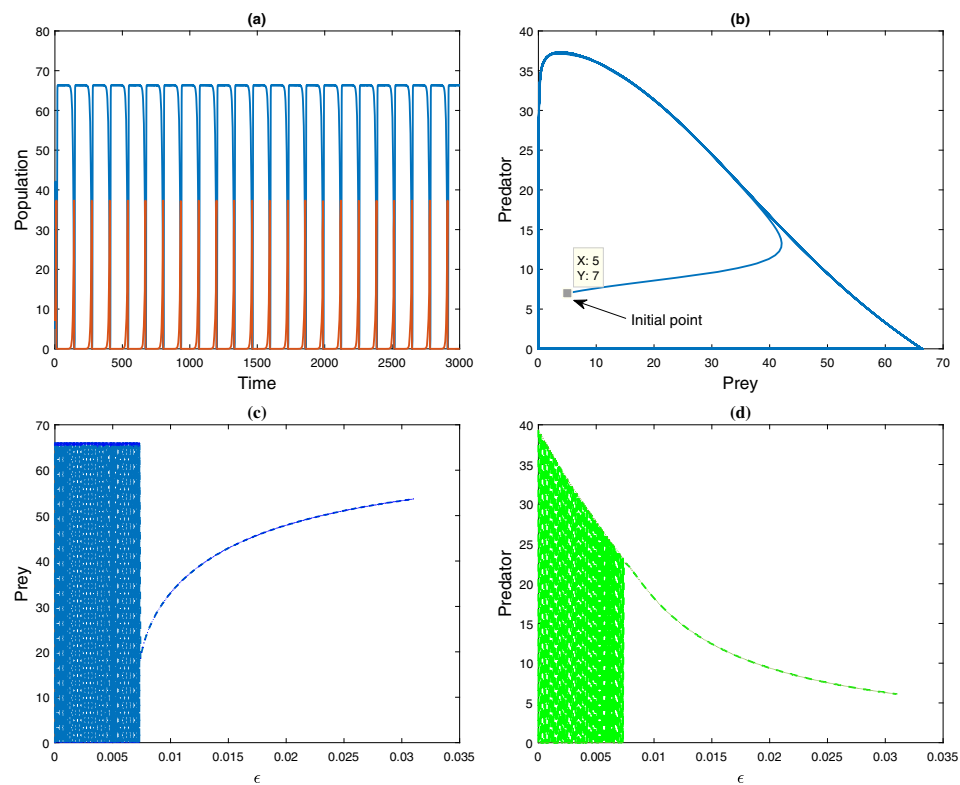


Fig. 10 The graphical representation of model (2.1) according to (6.1) with $\epsilon = 0.001$: **a** Periodic attractor. **b** 2D periodic attractor. **c** The bifurcation diagram of prey as a function of ϵ . **d** The bifurcation diagram of the predator as a function of ϵ

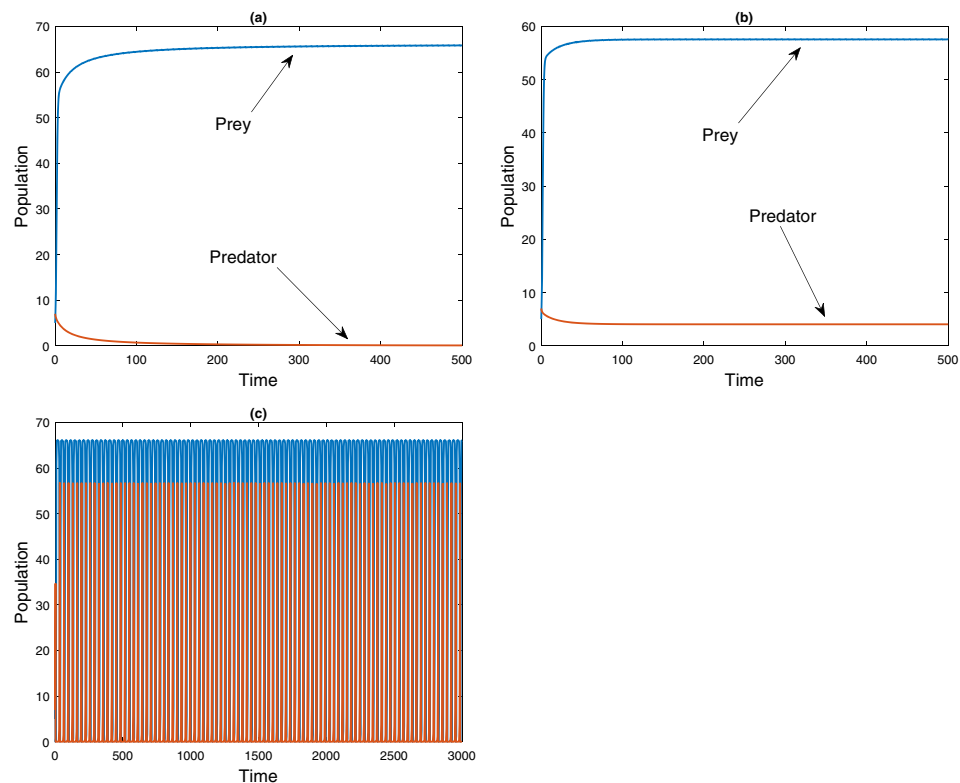


Fig. 11 The graphical representation of model (2.1) according to (6.1): **a** For $e \leq 0.01$, **b** for $0.01 < e \leq 0.5$, and **c** for $e \geq 0.6$

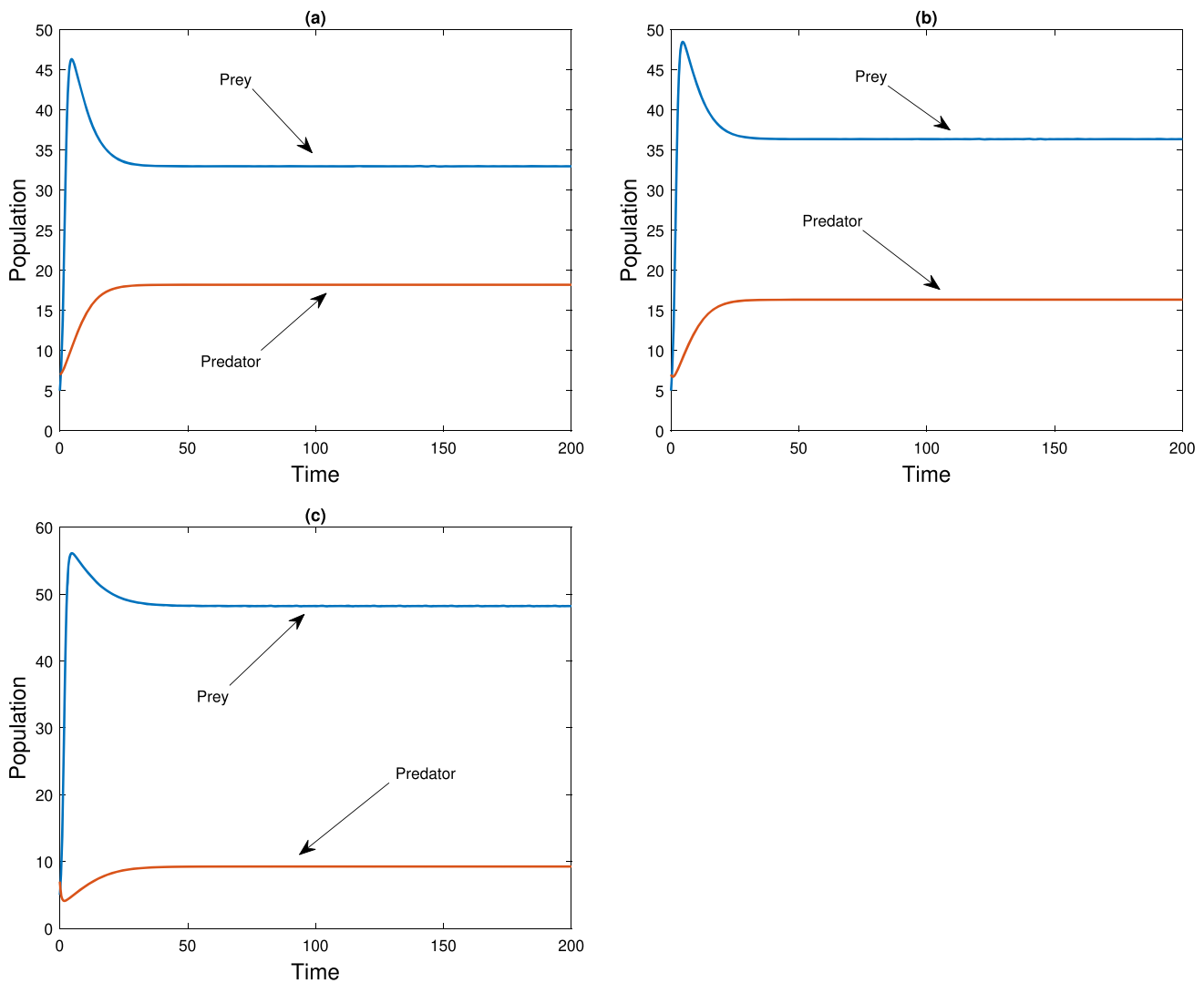


Fig. 12 The graphical representation of model (2.1) according to (6.1): **a** For $m = 0.1$, **b** for $m = 0.5$, and **c** for $m = 0.9$

The proposed model focused on factors that can influence its behavior, such as fear, immigration, refuge, and competition. It found that the model has three equilibrium points W_0 , W_1 , and W_2 . It was noted the EEP is in existence and is stable under condition ($r < a$), as well as the PFEP under condition (4.3) is stable, while the PIEP is stable when the condition (4.5) is held. Moreover, numerical simulation is performed to check each parameter's effect on the proposed model's dynamics and in the controlling and keeping of all species. Depending on numerical simulation, if the growth rate of prey decreases, then the PIEP becomes unstable, and the model solution converts to the periodic attractor. Also, the dynamical behavior of (PEP) converges to periodic attractors when the fear rate increases up to the vital value at which the system returns to the stable case as shown in the bifurcation diagram given in Fig. 7c and d. Additionally, consider the influence of the conversion rate of prey to predator e

on the dynamics of Model (2.1). According to the values of all parameters in equation (6.1), by setting different values of ($e = 0.01, 0.1, 0.6$) respectively, we get that, if $e < 0.01$, the solution of the model approaches W_1 . Also, when the value of e is between $0.01 < e \leq 0.5$, the dynamics of Model (2.1) converge to W_2 . But, if $e \geq 0.6$, the model exhibits a periodic solution at W_2 (see Fig. 11). Finally, Fig. 12 shows the quantitative effect of the refuge factor of prey on the dynamical behavior of Model (2.1). This effect occurs due to the existence of additional food. Our future research will investigate many ideas, such as the effect of time delay on the model, the memory effect by extending the model to a fractional one, and the impact of harvesting on the prey–predator dynamics.

Author Contribution R.Y.: conceptualization, methodology, formal analysis, validation, investigation, writing—original draft preparation, and writing—review. M.H.: writing—review, supervision, and editing. K.D.: conceptualization, visualization, methodology, writing—

review, and editing. A.M.: conceptualization, visualization, methodology, writing—review, and editing.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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